ZHONGMOU CHAO

Eric and Wendy Schmidt AI in Science Postdoc Fellow

Cornell University

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EDUCATION

Department of Chemical and Petroleum Engineering, University of Pittsburgh, PA, USA

Doctor of Philosophy in Chemical Engineering 12/2020

Department of Chemical Engineering, Columbia University, NY, USA

Master of Science in Chemical Engineering 02/2015

College of Chemistry and Chemical Engineering, Nanjing Tech University, China

Bachelor of Engineering in Chemical Engineering 07/2013

RESEARCH EXPERIENCE

Smith School of Chemical and Biomolecular Engineering, Cornell University, NY, USA

Eric and Wendy Schmidt AI in Science Postdoc Fellow, Daniel Lab and Frazier Lab 09/2023 – Present

- Developed an machine learning (ML) model for protein function prediction: better protein function prediction by modeling survivorship bias
- Leveraged Evo-PU to predict influenza viral protein functions and screen variants of concern
- Integrate ML, cell-free protein synthesis and bioelectronics to design drug delivery systems leveraging the functionalities of *de novo* viral proteins

Postdoc Associate, Daniel Lab 05/2021 – 08/2023

- Pioneered a bioelectronic sensor built on supported lipid bilayer (SLB)
- Developed viral infection-on-chip platform leveraging SLB-on-bioelectronics
- Integrated cell-free protein synthesis with bioelectronics for protein function studies

Department of Chemical and Petroleum Engineering, University of Pittsburgh, PA, USA

Graduate Research Assistant, Fullerton Lab 10/2015 – 04/2021

Dissertation Title: ***Silver Nanofilament Formation/Dissolution Dynamics Through a Polymer/Ionic liquid Composite by Direct-write***

- Investigated the impact of an ionic liquid on silver nanofilament formation kinetics through a solid polymer electrolyte
- Identified unique filament growth dynamics resulting from a competition between electrical double layer screening by the ionic liquid and silver electrodeposition
- Demonstrated how this newly identified growth mechanism could be useful for application in flexible electronics and neuromorphic computing

Department of Chemical Engineering, Columbia University, NY, USA

Graduate Research Assistant, West Lab 01/2014 - 06/2015

- Non-Photosynthetic Production of Biofuels from CO_2
- Production of Biofuels from Sulfur by A. Ferrooxidants Cells

PUBLICATIONS

1. **Chao, Z.**, Buathong, P., Selivanovitch, E., Daniel, S., and Frazier, P.. "Better Protein Function Prediction by Modeling Survivorship Bias." **submitted (2026)**. (//arxiv.org/abs/2605.06879)
2. Hattar, A., Alhammadi, J., Treiber, J., Hoven, D., **Chao, Z.**, Offenhaeusser, A., Daniel, S., Owens, R., Salleo, A., Santoro, F. and Pappa, A.M.. "Biomembranes in Hybrid Living Bioelectronics." *Nature Protocols*, **accepted (2026)**.
3. Treiber, J., **Chao, Z.**, McCoy, R., Lu, Z., Wheeler, A., Thiburce, Q., Daniel, S., Owens, R., Salleo, A.. "Reproducible High-Impedance Supported Lipid Bilayers on PEDOT:PSS." *Advanced Functional Materials*, e28703 **(2026)**. (//doi.org/10.1002/adfm.202528703)
4. Lu, Z., Treiber, J., Kallitsis, K., Selivanovitch, E., Wheeler, A., Lopez-Cavestany, M., **Chao, Z.**, Barron, S.L., Park, J.A., Hoven, D., Scheeder, A., Hass, B.M., Kaminski, C.F., Salleo, A., Pappa, A.M., Daniel, S. and Owens, R.. "Automated Label-Free Assay for Viral Detection and Inhibitor Screening via Biomembrane-Functionalized Microelectrode Arrays." *Advanced Materials*, e01985 **(2025)**. (//doi.org/10.1002/adma.202501985)
5. Su, Z., **Chao, Z.**, Jiang, V., Daniel, S. and Banta, S.. "Site Directed Mutagenesis of the Cyc2 Outer Membrane Protein from Acidithiobacillus ferrooxidans Reveals a Critical Role for Bound Iron Atoms in Extracellular Electron Transfer." *Small*, 2408837 (**2025**). (//doi.org/10.1002/smll.202408837)
6. **Chao, Z.**, Selivanovitch, E., Kallitsis, K., Lu, Z., Pachaury, A., Owens, R., and Daniel, S.. "Recreating the biological steps of viral infection on a cell-free bioelectronic platform to profile viral variants of concern." *Nature Communications*, 15, 5606 (**2024**). (//doi.org/10.1038/s41467-024-49415-6)
7. Liu, S.S., White, J.M., **Chao, Z.**, Li, R., Wen, S., Garza, A., Tang, W., Ma, X., Chen, P., Daniel, S. and Bates, F.S.. "A pseudo-surfactant chemical permeation enhancer to treat otitis media via sustained transtympanic delivery of antibiotics." *Advanced Healthcare Materials*, 13, 22, 2400457 **(2024)**. (//doi.org/10.1002/adhm.202400457)
8. Selivanovitch, E., Ostwalt, A., **Chao, Z.**, and Daniel, S.. "Emerging Designs and Applications for Biomembrane Biosensors." *Annual Review of Analytical Chemistry*, 17, 339-366 (**2024**). (//doi.org/10.1146/annurev-anchem-061622-042618)
9. Bint E Naser, S.F., Mohamed, Z.J., **Chao, Z.**, Bali, K., Owens, R.M. and Daniel, S.. "Gram-Positive Bacterial Membrane-Based Biosensor for Multimodal Investigation of Membrane–Antibiotic Interactions." *Biosensors*, 14, 45 (**2024**). (//doi.org/10.3390/bios14010045)
10. Bint E Naser, S.F., Su, H., Liu, H.Y., Manzer, Z.A., **Chao, Z.**, Roy, A., Pappa, A.M., Salleo, A., Owens, R.M. and Daniel, S.. "Detection of Ganglioside-Specific Toxin Binding with Biomembrane-Based Bioelectronic Sensors." *ACS Applied Bio Materials*, 4, 7942-7950 (**2021**). (//doi.org/10.1021/acsabm.1c00878)
11. **Chao, Z.**, Sezginel, K.B., Xu, K., Crouch, G.M., Gray, A.E., Wilmer, C.E., Bohn, P.W., Go, D.B. and Fullerton-Shirey, S.K.. "Silver Nanofilament Formation Dynamics in a Polymer-Ionic Liquid Thin Film by Direct Write." *Advanced Functional Materials*, 30, 1907950 (**2020**). (//doi.org/10.1002/adfm.201907950)
12. Xu, K., Liang, J., Woeppel, A., Bostian, M.E., Ding, H., **Chao, Z.**, McKone, J.R., Beckman, E.J. and Fullerton-Shirey, S.K.. "Electric Double-Layer Gating of Two-Dimensional Field-Effect

Transistors Using a Single-Ion Conductor." *ACS Applied Materials & Interfaces*, 11, 35879-35887 (**2019**). (//doi.org/10.1021/acsami.9b11526)

13. Han, D., Crouch, G.M., **Chao, Z.**, Fullerton-Shirey, S.K., Go, D.B. and Bohn, P.W.. "Nanopore-Templated Silver Nanoparticle Arrays Photopolymerized in Zero-Mode Waveguides." *Frontiers in Chemistry*, 7, 216, (**2019**). (//doi.org/10.3389/fchem.2019.00216)
14. **Chao, Z.**, Radka, B.P., Xu, K., Crouch, G.M., Han, D., Go, D.B., Bohn, P.W. and Fullerton-Shirey, S.K.. "Direct-Write Formation and Dissolution of Silver Nanofilaments in Ionic Liquid-Polymer Electrolyte Composites." *Small*, 14, 1802023 (**2018**). (//doi.org/10.1002/smll.201802023)
15. Guan, J., Berlinger, S.A., Li, X., **Chao, Z.**, e Silva, V.S., Banta, S. and West, A.C.. "Development of reactor configurations for an electrofuels platform utilizing genetically modified iron oxidizing bacteria for the reduction of CO2 to biochemicals." *Journal of Biotechnology*, 245, 21-27, (**2017**). (//doi.org/10.1016/j.jbiotec.2017.02.004)

CONFERENCE PRESENTAIONS

1. **Chao, Z.**; Buathong, P.; Selivanovitch, E.; Daniel, S.; Frazier, P. "Synergizing Evolution-Aware ML Model and Bioelectronics to Screen Viral Variants of Concern" 2025 AIChE 2025 Annual Meeting, Oral Presentation, Nov 2025, Boston, MA
2. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Recreating the Biological Steps of Viral Infection on a Cell-Free Bioelectronic Platform to Profile Viral Variants of Concern" 2024 CNF Annual Meeting, *Invited Talk*, Sep 2024, Ithaca, NY
3. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Decorating Bioelectronic Surface with Supported Lipid Bilayer to Study Viral Entry and More" MRS Fall 2023 Exhibition, Oral Presentation, Nov 2023, Boston, MA
4. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Electrical Sensing of SARS-CoV2 Entries Using Supported Lipid Bilayers" Asilomar Bioelectronics Symposium 2022, Poster Presentation, Sep 2022, Monterey, CA
5. **Chao, Z.**; Selivanovitch, E.; Daniel, S. "Electrical Sensing of SARS-CoV2 Entries Using Supported Lipid Bilayers" 2022 Bioanalytical Sensors Gordon Research Conference/ Seminar, Poster and Oral Presentation, Jun 2022, Newport, RI
6. **Chao, Z.**; Xu, K.; Fullerton-Shirey, S.K. "Ionic Liquid Regulated Silver Filament Formation in a Solid Polymer Electrolyte for Neuromorphic Applications" 2020 Virtual MRS Spring/Fall Meeting & Exhibit, Oral Presentation, Dec 2020
7. **Chao, Z.**; Fullerton-Shirey, S.K. "Formation/Dissolution of Silver Filaments Through an Ionic Liquid-Polymer Electrolyte Composite" MRS Fall Meeting & Exhibit, Oral Presentation, Dec 2019, Boston, MA
8. **Chao, Z.**; Gray, A.E.; Fullerton-Shirey, S.K. "Direct-Write Formation and Dissolution of Silver Nanofilaments in Ionic Liquid-Polymer Electrolyte Composites" Pittsburgh Quantum Institute 2019 Annual Conference, Poster Presentation, Apr 2019, Pittsburgh, PA
9. **Chao, Z.**; Crouch, G.; Han, D.; Bohn, P.; Go, D.; Fullerton-Shirey, S.K. "Formation/Dissolution of Silver Filaments Through an Ionic Liquid-Polymer Electrolyte Composite" AIChE Annual Meeting, Poster Presentation, Oct 2018, Pittsburgh, PA
10. **Chao, Z.**; Fullerton-Shirey, S.K. "Direct-Write Formation and Dissolution of Silver Nanofilaments in Ionic Liquid-Polymer Electrolyte Composites" Pittsburgh Quantum Institute at Pitt's Science 2018 Conference, Poster Presentation, Oct 2018, Pittsburgh, PA
11. **Chao, Z.**; Radka, B. P.; Fullerton-Shirey, S.K. "Formation/Dissolution of Conductive Silver Filaments through an Ionic Liquid/Polymer Electrolyte Thin Film" Pittsburgh Quantum Institute at Pitt's Science 2017 Conference, Poster Presentation, Oct 2017, Pittsburgh, PA

12. **Chao, Z.**; Crouch, G.; Han, D.; Bohn, P.; Go, G.; Fullerton-Shirey, S.K. "Formation/dissolution dynamics of conductive silver filaments through an ionic liquid/ polymer electrolyte thin film" 231^{st} ECS Meeting, Oral Presentation, May 2017, New Orleans, LA

AWARDS AND FELLOWSHIPS

Schmidt AI in Science, Cornell University, Ithaca, NY
The Eric and Wendy Schmidt AI in Science Postdoc Fellow 09/2023 to 09/2025

Smith School of Chemical and Biomolecular Engineering, Cornell University, Ithaca, NY
Smith Fellowship for Postdoctoral Innovation ($ 10,000 in Research Grant) 08/2022

Pittsburgh Quantum Institute, PA, USA
PQI 2019 Best Poster Award ($ 1,000 in Travel Funding) 05/2019

Department of Chemical and Petroleum Engineering, University of Pittsburgh, PA, USA
Outstanding Publication Award ($ 500 in Cash Prize) 12/2018

PPG Industries, Inc., PA, USA
Graduate Student Fellowship ($ 25,000) 03/2018

Celanese Chemical Industry Co., Ltd., China
Outstanding Engineer Award 01/2013

Shanghai Pujing Chemical Industry Co. Ltd., China
Most Improved Student Award of Fly Bursary 04/2012

College of Chemistry and Chemical Engineering, Nanjing Technology University, China
Outstanding Student Cadre 12/2012
Excellent Students in Three Fields 12/2011
Individual, second-class, third-class and first-class scholarship 09/2010, 01/2011, 09/2011 and 09/2012

TEACHING EXPERIENCE

Smith School of Chemical and Biomolecular Engineering, Cornell University, NY, USA
Guest Speaker on Departmental Work-in-Progress Seminar 10/2024

Department of Chemical and Petroleum Engineering, University of Pittsburgh, Pittsburgh, PA
Teaching Assistant for Undergraduate Transport Phenomenon 09/2017 to 12/2017, 09/2018 to 12/2018

OUTREACH ACTIVITIES

Cornell CBE Women Events 2022, Ithaca, NY 04/2022
- Volunteered the outreach activity organized by CBE WOMEN aiming at promoting interests and understandings in chemical engineering among local high school girls

Materials Science & Technology (MS&T) Outreach, Pittsburgh, PA 10/2017
- Designed and performed demo experiment for visiting middle school students near Pittsburgh area to observe polymer chain crystallization using iPad camera